

WATER LEVEL INDICATOR

Poster:



WATER LEVEL INDICATOR

SMART MONITORING • OVERFLOW PREVENTION • WATER CONSERVATION



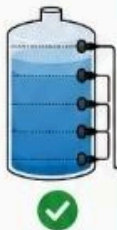
PROBLEM

Water tanks often overflow or run empty due to lack of real-time monitoring, leading to water wastage and manual checking.

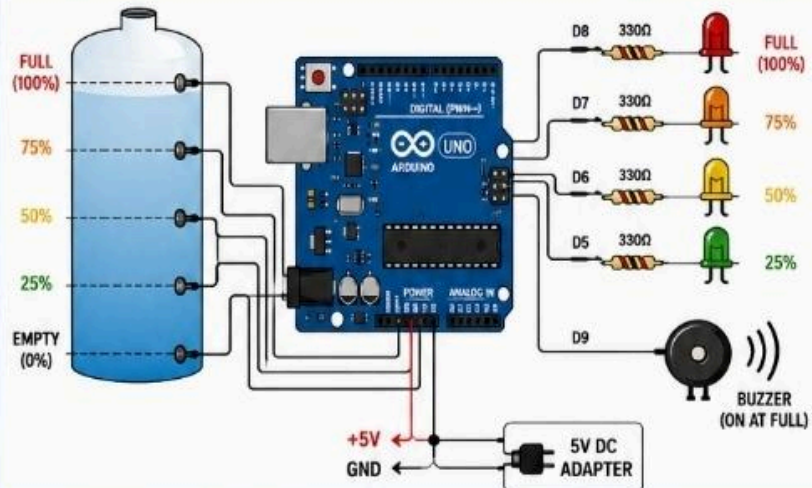


SOLUTION

This Water Level Indicator monitors the water level in real-time and shows its status using LEDs. It alerts the user with a buzzer when the tank is full.



SYSTEM OVERVIEW



FEATURES

- ✓ 5 Level Indication (Empty to Full)
- ✓ LED Indicators for Easy Monitoring
- ✓ Buzzer Alert at Full Level
- ✓ Simple, Low Cost & Easy to Build
- ✓ Helps in Water Conservation



SAVE WATER
SAVE LIFE

HOW IT WORKS

Probes are placed at different levels inside the tank. When water touches a probe, it conducts electricity and sends a HIGH signal to the Arduino analog pin. Arduino detects the highest level reached and turns ON the corresponding LED.

When the water reaches the FULL level, the buzzer turns ON to alert the user.

LEVEL	PROBE (ANALOG)	LED STATUS	BUZZER
FULL (100%)	A0	Red (D8)	ON
75%	A1	Orange (D7)	OFF
50%	A2	Yellow (D6)	OFF
25%	A3	Green (D5)	OFF
EMPTY (0%)	A4	All OFF	OFF

APPLICATIONS



Home Water Tanks



Residential Apartments



Industries & Factories



Agriculture & Irrigation



Water Storage Systems

COMPONENTS USED

- Arduino UNO
- 5 x Probes (Steel/Conductive)
- 4 x LEDs (Red, Orange, Yellow, Green)
- 1 x Active Buzzer (5V)
- 4 x 330Ω Resistors
- Jumper Wires
- 5V DC Adapter
- Tank / Container



CIRCUIT DIAGRAM

(Refer to the diagram for connections)



SCAN TO VIEW CIRCUIT

BENEFITS

- ✓ Prevents water overflow
- ✓ Saves water and reduces wastage
- ✓ Reduces manual effort
- ✓ Low power consumption
- ✓ Easy installation and maintenance



CONCLUSION

The Water Level Indicator is a simple and effective solution for monitoring water levels in tanks. It helps in preventing overflow, saves water, and reduces manual checking. This project can be further enhanced by adding IoT for remote monitoring.

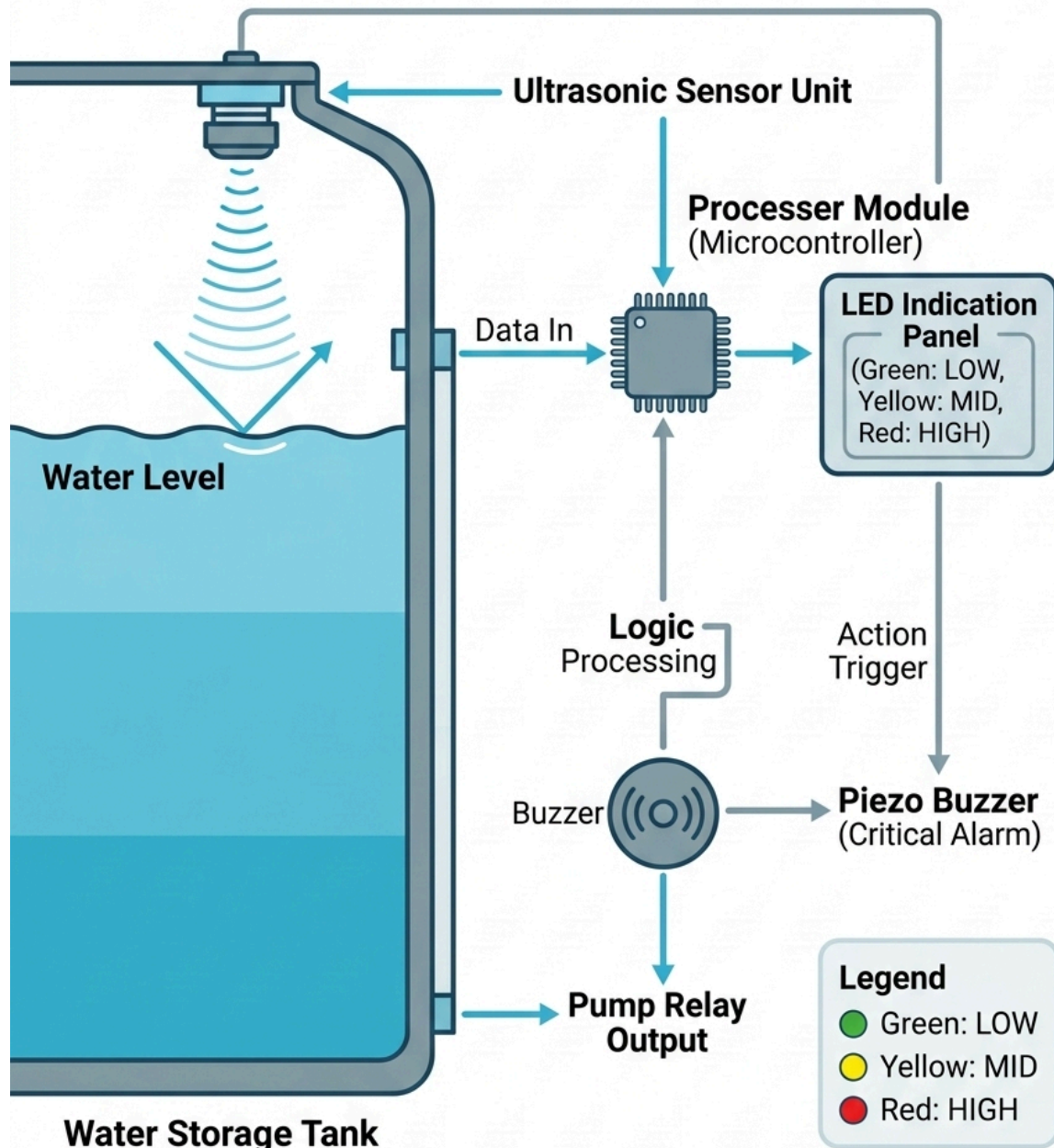
“ Every drop counts.
Let's build a smarter
and sustainable future.
”



CONSERVE WATER
SECURE TOMORROW

Visual:

Smart Water Level Indicator System



Circuit diagram:

